

=====

SECTION 1: HYDRAULIC & FOULING OPERATIONAL THRESHOLDS

1.1 Pressure Drop (Delta-P) Limits:

- Clean Baseline Shell Delta-P: 0.70 - 0.90 bar.
- Advisory Alert Threshold: Delta-P > 1.50 bar (Plan off-line hydro-jetting).
- Critical Fouling Limit: Delta-P > 2.20 bar (Mandatory bypass diversion within 72 hrs).

SECTION 2: THERMAL DEGRADATION THRESHOLDS

2.1 Overall Heat Transfer Coefficient Degradation (U/U_design):

- **Acceptable: > 75% of design duty.**
- Marginal / Clean Scheduling Required: 60% to 75%.
- Severe Degradation / Unacceptable Loss: < 55% of design heat transfer.

SECTION 3: HYDROSTATIC PRESSURE TESTING & LEAK DETECTION

3.1 Shell-Side Test Pressure: 1.5 x Design Pressure = 37.5 bar gauge.

3.2 Tube-Side Test Pressure: 1.5 x Design Pressure = 28.0 bar gauge.

3.3 Holding Time: Minimum 60 minutes with zero observable pressure decay.

3.4 Tube Plugging Limit: Maximum allowable plugged tubes is 10% of total count (42 tubes).

SECTION 4: CHEMICAL CLEANING SAFETY PROTOCOL

4.1 Ensure full gas freeing and LEL < 0% before opening channel head cover.

4.2 Neutralize acidic residues with 2% sodium carbonate wash prior to water lance entry.